

Progress report, 29 September 2026: the decoder's moments from operator size — exact sum rules, two correlated BPS spaces, and chord pictures toward a gravity dual

syk_gravity project

Drafted by Claude for J. Chryssanthacopoulos; code state: commit 774934a + working tree

Abstract

Yesterday's checkpoint showed that the variance of the uplift decoder $O = P(1 - n_i)P$ of one-flavor $\mathcal{N}=2$ SYK is reproduced to about 2% by the finite- λ super-chord theory. It also showed that, in that theory, the Haar/Wachter value is exactly the contribution of the zero-length supersymmetric wormhole. This report takes the next steps. It keeps only results that survived a claim-by-claim audit of the working note.

- **Two correlated BPS spaces.** The decoder spectrum is the principal-angle law between the BPS space B and $B' = UB$, where $U = (-1)^{n_i}$. B' is the BPS space of the same model with the new mode's couplings reversed. In chord language the zero-temperature boundary splits into P - and P' -arcs, and chords joining the two kinds carry a weight x .
- **The variance is operator size.** The SYK coupling ensemble is $U(N)$ -invariant. As a result, the decoder variance is an *exact finite- N functional of the operator-size spectrum w_k of the BPS projector*:

$$r = 1 - \frac{4 \langle k(N+1-k) \rangle_w}{N(N+1)} \quad (\text{half filling}), \quad w_0 = a \text{ exactly.}$$

This is the exact counterpart of the wormhole identity: the identity (size-0) component of P plays the zero-length wormhole. It reproduces SYK to 0.1–0.8%.

- **Freeness is flat transmission.** Two independent SYK BPS spaces are *not* free relative to each other. Their fourth moment is predicted exactly from single-model size data, and the prediction matches pair measurements to 0.01–0.2%.
- **The fourth moment.** The decoder's T_4 splits exactly into a part fixed by the variance, a part fixed by the size spectrum, and a four-point remainder R ($\approx 21\%$ of the excess at $N' = 13$ – 14), which remains open.

The exact identities are kinematics (representation theory of $U(N)$). The dynamics, and hence any gravity dual, enters only through the size spectrum and its multi-copy generalizations. Tags: [\[derived\]](#), [\[derived + verified\]](#), [\[numerical\]](#), [\[interpretation\]](#), [\[conjecture\]](#).

Contents

1 Summary of results	2
2 Setup and recap	3

3	The decoder as the overlap of two correlated BPS spaces	4
3.1	Parity form [derived + verified]	4
3.2	Chord picture: arcs and the Z_2 rule [derived]	4
3.3	Tuning x at fixed BPS space: the parity family [numerical]	4
4	Operator size of the BPS projector: an exact variance sum rule	5
4.1	$U(N)$ invariance [derived]	5
4.2	Size sectors [derived + verified]	6
4.3	The sum rule and its closed form for the decoder [derived + verified]	6
4.4	Only even sizes at half filling [derived + verified] (sketch)	7
4.5	SYK numbers [numerical]	7
4.6	Comparison with the super-chord theory [numerical] + [interpretation]	8
5	Transmission, and why two BPS spaces are not free	9
5.1	The transmission spectrum [derived]	9
5.2	Transmission is a universal function of size [derived + verified]	9
5.3	Exact sum rules; freeness is flat transmission [derived + verified]	10
5.4	Two independent SYK models: prediction vs measurement [numerical]	10
6	The decoder's fourth moment	10
6.1	Exact decomposition [derived + verified]	10
6.2	SYK numbers [numerical]	11
6.3	What is known about R	11
7	Toward a gravity dual of the decoder	12
7.1	Kinematics versus dynamics	12
7.2	Dictionary so far	13
7.3	Next computations	13
8	Open questions and dubious points	13
A	Derivation details	14
A.1	The reflection's characters	14
A.2	The transmission matrix	14
A.3	Haar transmission	15
B	Reproducibility	15

1 Summary of results

- [derived + verified] **Parity form.** $1 - n_i = \frac{1}{2}(1 + U)$, $U = (-1)^{n_i}$, so the decoder is $\frac{1}{2}(1 + PUP)$. With $P' = UPU$, $(PUP)^2 = PP'P$. P' is the BPS projector of $Q' = UQU$: the same supercharge with every coupling containing mode i sign-flipped (§3).
- [derived] **Z_2 chord rule.** A Q -chord whose endpoints are separated by an odd number of U -insertions picks up $(-1)^{[i \in I]}$, i.e. weight x on average. Chords joining arcs of the same kind get 1. This is exact per chord; independence between chords is the double-scaled approximation (§3.2, Fig. 1–2).

3. **[numerical] Tuning x at fixed P .** Parity projectors on s modes tune x from 0.57 ($s = 1$, the decoder) to exactly 0 ($s = N'/2$).
 - The second moment becomes free ($r \rightarrow a$).
 - The fourth and sixth moments do not (+5.6%, +10% at $N' = 14$, growing with N') (§3.3).
4. **[derived + verified] Exact variance sum rule.**
 - $E[P] = a\mathbb{1}$, and $E[T_2] = \sum_k w_k \hat{\chi}_k(U)$, with $w_0 = a$ exactly.
 - For the single-mode decoder, $\hat{\chi}_k = 1 - 4k(N+1-k)/(N(N+1))$, which gives the boxed formula above.
 - It agrees with the SYK campaign to 0.1–0.8 % for $N' = 8 \dots 14$ (§4).
5. **[derived + verified] Only even sizes at half filling.** Odd-size weights are $\leq 2 \times 10^{-28}$, i.e. exactly zero. A particle–hole argument (sketch) explains it; the same symmetry gives the known $\lambda \leftrightarrow 1 - \lambda$ pairing.
6. **[numerical] Size $\leftrightarrow$ wormhole length (empirical).** w_{2n} tracks the super-chord wormhole-length distribution P_n to 10–20 %, and the characters track the matter weights x^{2n} (§4.6).
7. **[derived + verified] Transmission.**
 - $X \mapsto E[XPX]$ acts on size- k operators as a scalar ϕ_k , which is a universal linear function of w .
 - Haar: $\phi_{k \geq 1} = a^2$ (flat). SYK: $\phi_k > a^2$ at small k .
 - Two independent models: $E\tau(P_1 P_2 P_1 P_2) = \sum_k \phi_k w_k$, exactly. SYK is +0.6, +2.0, +4.3% above free at $N' = 10, 12, 14$, and the prediction matches measurement to $\leq 0.2\%$ (§5).
8. **[derived + verified] Decoder fourth moment.** $T_4 = a^2 + 2a(T_2 - a) + S + R$ exactly, with $S = \sum_{k \geq 1} \phi_k w_k$. At $N' = 13\text{--}14$, 74 % / 5 % / 21 % of the excess over free come from the variance term, the size-spectrum term and the remainder R (§6).
9. **[derived] Correction to yesterday’s conjecture.** “The free law is the zero-length truncation” holds for m_2 only. Already for Haar P , the free T_4 receives $a^2(1 - a)$ from nonzero sizes. The correct characterization is: *freeness = flat transmission*.

2 Setup and recap

Model and decoder. One-flavor $\mathcal{N}=2$ SYK on N' complex fermions with $Q = \sum_{|I|=p} C_I \psi_I$ ($p = 3$) and i.i.d. complex Gaussian couplings; $H = \{Q, Q^\dagger\}$.

- Fermion-number sector $F = \lfloor N'/2 \rfloor$ (half filling at even N'), the Fock space Λ^F of dimension $D = \binom{N'}{F}$. $B \subset \Lambda^F$ is the BPS space, $d = \dim B$, $a = d/D$. P always denotes the BPS projector, and $\tau = \text{Tr}_B / d$. In §§4–5 and Appendix A we write $N \equiv N'$. (The code and earlier notes call the sector P .)
- Decoder: $O = P\Pi P$ with $\Pi = 1 - n_i$ (i the new mode; all modes are equivalent in distribution). Eigenvalues $\lambda \in [0, 1]$; $b = 1 - F/N'$.
- Rigidity: $r = \text{Var}(\lambda)/[b(1 - b)]$. Haar/Wachter predicts $r = a$.

Yesterday's result [numerical] + [literature]. The Boruch–Lin–Yan (BLY) [2] double-scaled two-point function, with $\lambda = 2p^2/N'$ and $\Delta = 1/p$, gives $r = \sum_n P_n x^{2n}$. Here P_n is the chord-number (wormhole-length) distribution of the supersymmetric Hartle–Hawking state, $x = q^\Delta$, and $P_0 = a_\lambda$ is the double-scaled BPS fraction. It matches SYK to $\approx 2\%$. The Haar value is the $n = 0$ term. That identity was exact only within the double-scaled theory, and a_λ exceeds the exact a by 2–8 % at $N' = 8\text{--}14$ (Table 3). Section 4 removes both limitations.

3 The decoder as the overlap of two correlated BPS spaces

3.1 Parity form [derived + verified]

Since n_i is a projector, $\Pi = 1 - n_i = \frac{1}{2}(1 + U)$ with $U = (-1)^{n_i}$, a unitary involution. So $P\Pi P = \frac{1}{2}(P + PUP)$, and on B

$$\lambda = \frac{1}{2}(1 + \mu), \quad \mu \in \text{spec}(PUP|_B) \subset [-1, 1], \quad T_m \equiv \tau((PUP)^m) = E[\mu^m]. \quad (1)$$

Using $U^2 = 1$,

$$(PUP)^2 = P(UPU)P = PP'P, \quad P' \equiv UPU. \quad (2)$$

P' is the BPS projector of $Q' = UQU = \sum_I (-1)^{[i \in I]} C_I \psi_I$, the model with all couplings involving mode i reversed. Hence μ^2 are the squared cosines of the principal angles between B and $B' = UB$. In uplift language (Chryssanthacopoulos–Vegh [1]): writing a BPS state as $\Psi = a + b \wedge e_i$, one has $\langle \Psi | U \Psi \rangle = \|a\|^2 - \|b\|^2 = 2\lambda - 1$. So the decoder measures how much the BPS space rotates when the new mode's couplings are reversed. At half filling the spectrum is symmetric ($T_{\text{odd}} = 0$, verified to 10^{-16} ; §4.4), so the principal-angle law determines the decoder law completely.

3.2 Chord picture: arcs and the $\mathbb{Z}_2$ rule [derived]

For a supercharge term, $U\psi_I U = (-1)^{[i \in I]} \psi_I$. In a trace of a $Q, Q^\dagger$ word with an *even* number m of U insertions, move all U 's together ($U^m = 1$). Each supercharge factor picks up $(-1)^{[i \in I]}$ once for every U it passes. A Q -chord (a Wick pair $C_I C_I^*$) therefore carries $(-1)^{[i \in I] \cdot \#}$, where $\#$ is the number of U 's separating its endpoints. Averaged over its random index set I :

$$\text{weight} = \begin{cases} x, & \# \text{ odd} \\ 1, & \# \text{ even} \end{cases}, \quad x = 1 - \frac{2p}{N'} \text{ (exact, per chord)}, \quad x \simeq e^{-2p/N'} = q^{1/p} \text{ (double-scaled)}, \quad (3)$$

since $i \in I$ with probability p/N' ; here $q = e^{-\lambda}$, $\lambda = 2p^2/N'$. Label the m zero-temperature arcs alternately P and P' . A chord joining a P -arc to a P' -arc gets x ; a chord joining arcs of the same kind gets 1. Figure 1 shows $m = 2$, which reproduces BLY's horizontal matter chord and $r = \sum_n P_n x^{2n}$. Figure 2 shows $m = 4$. There a chord spanning two opposite P -arcs gets weight 1, not x^2 , so the higher moments are *not* given by the Gaussian-matter (q-Gaussian) rule. That the per-chord averages multiply is the double-scaled approximation: at finite N , different chords share site indices.

3.3 Tuning x at fixed BPS space: the parity family [numerical]

Replace U by $U_S = (-1)^{N_S}$ with $|S| = s$ and $S \ni i$. Then $U_S \psi_I U_S = (-1)^{|I \cap S|} \psi_I$, and

$$x_s = \sum_k (-1)^k \binom{s}{k} \binom{N' - s}{p - k} / \binom{N'}{p} \quad \text{(exact per chord)}. \quad (4)$$

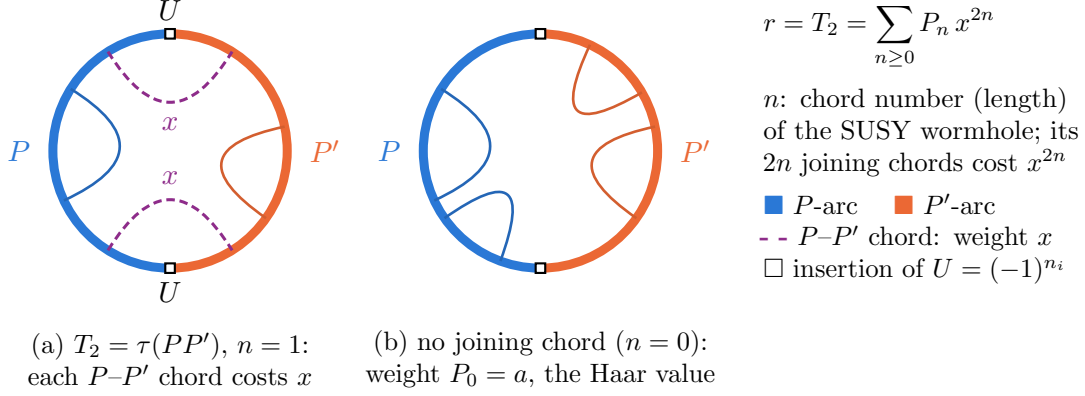


Figure 1: [derived] The second moment as a two-arc chord diagram at zero temperature. The two U insertions split the boundary into a P -arc and a P' -arc. Chords with both ends on one arc cost 1. Chords joining the arcs cost x each. Summing over diagrams gives $\sum_n P_n x^{2n}$ (BLY eqs. 4.5–4.8, with yesterday’s identification $\lambda = 2p^2/N'$, $\Delta = 1/p$). The $n = 0$ class (b) is the zero-length wormhole, whose weight P_0 is the BPS fraction, i.e. the free value.

For odd p , $|I \cap S^c| = p - |I \cap S|$ gives $x_{N'-s} = -x_s$, so $x = 0$ *exactly* at $s = N'/2$. (With the charge fixed, $U_{S^c} = \pm U_S$ is the same probe.) The BPS projector is the same for every s ; only the probe changes. Figure 3 and Table 1 give the results.

s	$N' = 14$, 4 realizations					$x \approx 0$ across sizes	
	x_s	r/a	T_4/T_4^{free}	T_6/T_6^{free}	KS to Wachter	N'	T_4/T_4^{free} at $s = \lfloor N'/2 \rfloor$
1 (decoder)	+0.571	1.518	1.887 ± 0.018	2.21	0.145	10	1.014 ± 0.007
2	+0.275	1.208	1.359 ± 0.024	1.49	0.080	12	1.025 ± 0.004
3	+0.088	1.062	1.144 ± 0.024	1.22	0.035	13	1.032 ± 0.010
4	-0.011	1.007	1.056 ± 0.014	1.11	0.026	14	1.056 ± 0.002
7	0	1.010	1.056 ± 0.002	1.10	0.021	Haar	1.000 ± 0.001

Table 1: [numerical] Moments relative to free compression (Wachter(a, b)). The KS distance of Haar controls is 0.002. $T_4^{\text{free}} = 2a^2 - a^3$ and $T_6^{\text{free}} = 5a^3 - 6a^4 + 2a^5$ at $b = \frac{1}{2}$ (free cumulants of a symmetric ± 1 variable: $\kappa_2 = 1$, $\kappa_4 = -1$, $\kappa_6 = 2$).

Findings.

- **The variance becomes free** as the matter weight is removed. The small residual at $x = 0$ ($r/a = 1.010 \pm 0.002$ at $N' = 14$) is accounted for at the 2σ level by §4 (predicted 1.006).
- **The fourth moment does not.** An x -independent excess remains and grows with N' . Section 5 identifies its origin: two different SYK BPS spaces are not mutually free.

4 Operator size of the BPS projector: an exact variance sum rule

4.1 $U(N)$ invariance [derived]

The couplings are i.i.d. standard complex Gaussians in the orthonormal basis $e_{i_1} \wedge \cdots \wedge e_{i_p}$ of $\Lambda^p \mathbb{C}^N$. The induced action of $g \in U(N)$ (single-particle rotations) is unitary, so the coupling law is

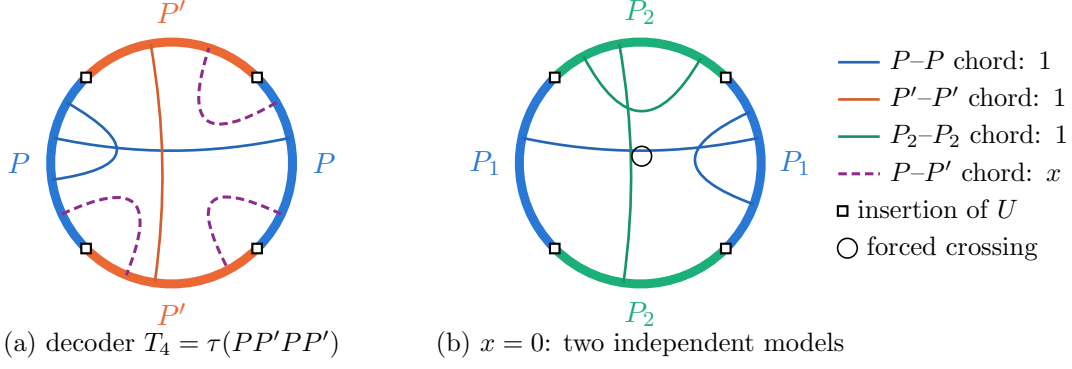


Figure 2: [derived] + [interpretation] Four arcs. (a) The decoder’s fourth moment $T_4 = \tau(PP'PP')$. A chord spanning the two P -arcs (or the two P' -arcs) separates an even number of U ’s and costs 1, not x^2 : the Z_2 rule differs from Gaussian matter. (b) At $x = 0$ only same-kind chords survive, as for two independent SYK models. The two “across” chords must then cross (circled), and their crossing weight comes from shared site indices. If such crossings were forbidden (crossing weight 0, as for freely independent q -Gaussians), the two BPS spaces would be free. In SYK they are not: $E \tau(P_1 P_2 P_1 P_2) - T_4^{\text{free}} = \sum_{k \geq 1} \delta \phi_k w_k > 0$ (§5).

$U(N)$ -invariant. $Q = \omega \wedge$ and $Q^\dagger$ are covariant, so $P(g\omega) = \rho(g)P(\omega)\rho(g)^\dagger$ and the BPS-projector ensemble is invariant under conjugation. Since Λ^F is irreducible, Schur’s lemma gives

$$E[P] = a \mathbb{1} \quad \text{exactly.} \quad (5)$$

4.2 Size sectors [derived + verified]

Under $U(N)$ conjugation, $\text{End}(\Lambda^F) = \Lambda^F \otimes (\Lambda^F)^*$ decomposes multiplicity-free,

$$\text{End}(\Lambda^F) = \bigoplus_{k=0}^{\min(F, N-F)} V_k, \quad \dim V_k = \binom{N}{k}^2 - \binom{N}{k-1}^2. \quad (6)$$

Here V_k consists of the traceless “ k -body” operators (highest weight $(1^k, 0, \dots, 0, (-1)^k)$), and the adjoint Casimir $C(X) = \sum_{ab} [E_{ab}, [E_{ba}, X]]$, with $E_{ab} = \bar{\psi}_a \psi_b$, acts on V_k as $2k(N+1-k)$. Write $P = \sum_k P^{(k)}$ and define the **size spectrum**

$$w_k = \frac{\|P^{(k)}\|_F^2}{d}, \quad \sum_k w_k = 1, \quad \boxed{w_0 = a \text{ exactly, per realization}}, \quad (7)$$

since $P^{(0)} = (\text{Tr } P/D) \mathbb{1} = a \mathbb{1}$. (Verified: full diagonalization of the Casimir superoperator at $N = 6$ reproduces the spectrum, the multiplicities and the parity characters; the size components reconstruct P with residual $< 10^{-10}$ at $N = 8$; `tests/test_size_decomposition.py`.)

4.3 The sum rule and its closed form for the decoder [derived + verified]

Any $g \in U(N)$ preserves every V_k , so $\tau(PgPg^\dagger) = \frac{1}{d} \sum_k \langle P^{(k)}, \text{Ad}_g P^{(k)} \rangle$ with no cross terms. By invariance, $E[|P^{(k)}\rangle\rangle\langle\langle P^{(k)}|] \propto \mathbb{1}_{V_k}$ (Schur), hence

$$E[\tau(PgPg^\dagger)] = \sum_k w_k \hat{\chi}_k(g), \quad \hat{\chi}_k(g) = \frac{|\chi_{\Lambda^k}(g)|^2 - |\chi_{\Lambda^{k-1}}(g)|^2}{\dim V_k}. \quad (8)$$

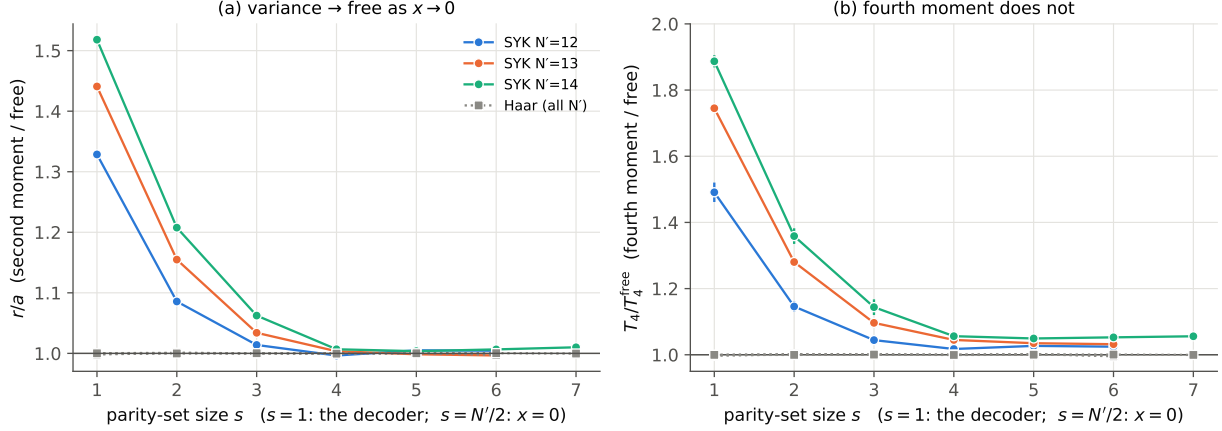


Figure 3: [numerical] Parity family at $N' = 12, 13, 14$ (4 SYK realizations each; Haar controls dotted). Exact per-chord weights at $N' = 14$: $x_s = 0.571, 0.275, 0.088, -0.011, -0.044, -0.033, 0$ for $s = 1 \dots 7$. (a) The rigidity r/a returns to the free value as $x \rightarrow 0$. (b) The fourth moment does not: an x -independent excess remains. The free baseline is the Wachter(a, b) law (with $b \neq \frac{1}{2}$ at odd N').

Equation (8) is exact at finite N , for any charge sector and any group element. For parities, $\chi_{\Lambda^k}(U_S)$ is a Krawtchouk polynomial. For the decoder ($g = U$, a reflection), the character is a linear function of the Casimir (Appendix A.1):

$$\hat{\chi}_k(U) = 1 - \frac{4k(N+1-k)}{N(N+1)} \implies E[T_2] = 1 - \frac{4 \langle k(N+1-k) \rangle_w}{N(N+1)} = a + \sum_{k \geq 1} w_k \hat{\chi}_k(U) \quad (9)$$

At half filling $E[T_2] = r$.

- **The decoder variance measures the mean operator size (adjoint Casimir) of the BPS projector.**
- The identity component contributes exactly a , the Haar value. The deviation from freeness is P 's weight on nonzero sizes, weighted by the probe's characters.
- **Haar check [derived].** For Haar P , $w_k = (1-a) \dim V_k / (D^2 - 1)$ ($k \geq 1$). Since $\sum_k \dim V_k \hat{\chi}_k = |\text{Tr } U|^2$, this gives $E[T_2] = (aD^2 - 1) / (D^2 - 1)$ at half filling, the exact Weingarten value.

4.4 Only even sizes at half filling [derived + verified] (sketch)

Particle-hole conjugation V ($\psi \leftrightarrow \bar{\psi}$) maps $\Lambda^{N/2}$ to itself and Q_ω to $Q_\omega^\dagger$. Hence $VB(\omega) = B(\bar{\omega}) = \overline{B(\omega)}$, i.e. $VPV^\dagger = P^T$. The map $X \mapsto (VXV^\dagger)^T$ acts on V_k as $(-1)^k$, so $P^{(k)} = 0$ for odd k . (Signs are not tracked in detail. Numerically, odd weights are $\leq 2 \times 10^{-28}$ at $N' = 8-14$, and nonzero at odd N' .) The same map sends $n_i \mapsto 1 - n_i$, which gives $T_{\text{odd}} = 0$ and the pairing $\lambda \leftrightarrow 1 - \lambda$.

4.5 SYK numbers [numerical]

Table 2 tests eq. (9) against the independent chord-invariant campaign of 28 September. It has no free parameter: the size weights come from a separate set of realizations, and the characters are exact. Figure 4a shows where the weight sits.

N'	$w_0 (= a)$	w_2	w_4	w_6	$\langle k \rangle_w$	$E[T_2]$, eq. (9)	SYK T_2	diff.
8	0.7714	0.2128	0.0158	–	0.49	0.8170	0.8186 ± 0.0008	–0.2%
10	0.6429	0.2997	0.0574	–	0.83	0.7454	0.7463 ± 0.0007	–0.1%
12	0.5260	0.3486	0.1119	0.0135	1.23	0.6855	0.6874 ± 0.0009	–0.3%
13*	0.4248	0.3398	0.1435	0.0318	1.64	0.6176	0.6193 ± 0.0018	–0.3%
14	0.4248	0.3706	0.1629	0.0416	1.64	0.6371	0.6420 ± 0.0037	–0.8%
14, Haar	0.4248	0.0004	0.0425	0.2448	3.33	0.4248	$0.4249^\dagger$	

Table 2: [numerical] Size spectrum (4 SYK realizations per N' , seeds 0–3) and the parameter-free sum rule against the mode-averaged chord-invariant campaign of 28 September (60, 40, 16, 10, 4 realizations at $N' = 8, 10, 12, 13, 14$). At even N' , $T_2 = r$, and odd sizes vanish. *At odd N' the odd sizes are nonzero ($w_{1,3,5} = 0.0146, 0.0215, 0.0239$), and the campaign $r = 0.6171 \pm 0.0018$ is converted to $T_2 = r(1 - c^2) + c^2$ with $c = 2b - 1 = 1/13$. The Haar row lists w_2, w_4, w_6 (its weight sits mostly at $k = 5-7$); † direct decoder measurement on that projector.

4.6 Comparison with the super-chord theory [numerical] + [interpretation]

Equation (9) and BLY’s $r = \sum_n P_n x^{2n}$ have the same structure. Table 3 compares the ingredients term by term at $k = 2n$.

N'	n	w_{2n} (exact size weight)	P_n (BLY, $\lambda = 18/N'$)	$\hat{\chi}_{2n}(U)$ (exact)	x^{2n} ($x = e^{-2p/N'}$)
12	0	0.5260	0.5587	1	1
12	1	0.3486	0.3209	0.4359	0.3679
12	2	0.1119	0.0922	0.0769	0.1353
12	3	0.0135	0.0218	–0.0769	0.0498
14	0	0.4248	0.4588	1	1
14	1	0.3706	0.3506	0.5048	0.4244
14	2	0.1629	0.1339	0.1619	0.1801
14	3	0.0416	0.0407	–0.0286	0.0764

Table 3: [numerical] The exact size spectrum tracks the chord-length distribution, and the exact characters track the double-scaled matter weights, each to 10–20%. The sums agree to about 2%, which is consistent with partial cancellation of the termwise differences.

[interpretation] This is the exact finite- N content of yesterday’s wormhole identity:

- the zero-length wormhole $\leftrightarrow$ the identity component $P^{(0)} = a\mathbb{1}$ (weight $w_0 = a$ exactly, so the a_λ -vs- a discrepancy disappears);
- a wormhole of n chords $\leftrightarrow$ the size- $2n$ component;
- the matter propagator $x^{2n} \leftrightarrow$ the normalized character of the probe on that sector.

The pairing $n \leftrightarrow k/2$ is an empirical match, not a derivation (§8).

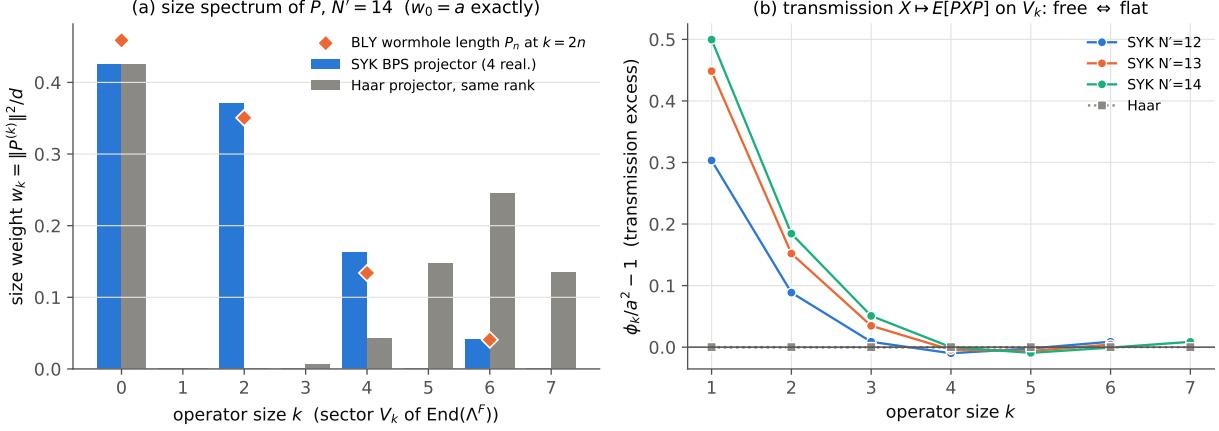


Figure 4: (a) [numerical] Size spectrum of the BPS projector at $N' = 14$. SYK puts its non-identity weight on small, even sizes; a Haar projector spreads it $\propto \dim V_k$. Orange diamonds: BLY's wormhole-length distribution P_n at $\lambda = 18/14$, placed at $k = 2n$ (empirical identification, §4.6). (b) [derived + verified] Transmission excess $\phi_k/a^2 - 1$ (§5). Haar is flat (0). SYK transmits small operators more than freeness allows, increasingly with N' , with a small compensating deficit at intermediate k (sum rule (13)).

5 Transmission, and why two BPS spaces are not free

5.1 The transmission spectrum [derived]

Define $\Phi(X) = E[XPXP]$. By invariance, Φ commutes with Ad_g on $\text{End}(\Lambda^F)$, so by Schur it acts on each V_k as a scalar,

$$\Phi|_{V_k} = \phi_k, \quad E \text{Tr}(XPXP^\dagger) = \sum_k \phi_k \|X^{(k)}\|^2 \quad \text{for any } X. \quad (10)$$

ϕ_k is the *transmission* of a size- k operator through the BPS projection. For two independent models, $E \text{Tr}(P_1 P_2 P_1 P_2) = E \text{Tr}(P_1 \Phi(P_1))$, so

$$E \tau(P_1 P_2 P_1 P_2) = \sum_k \phi_k w_k \quad (11)$$

5.2 Transmission is a universal function of size [derived + verified]

The superoperator $X \mapsto XPXP$, i.e. $P \otimes P^T$, is a *realignment* of $|P\rangle\rangle\langle\langle P|$, the object whose twirl gives w_k . Realignment is $U(N)$ -equivariant, so it maps the commutant $\text{span}\{\Pi_k\}$ to itself. Hence, per realization (for the $U(N)$ -twirled transmissions),

$$\phi_j = \sum_k \mathcal{A}_{jk} \frac{dw_k}{\dim V_k}, \quad (12)$$

with a universal matrix $\mathcal{A}(N, F)$. We compute $\mathcal{A}$ on the maximal torus, where the size- k part of a diagonal operator is its projection onto the k -th eigenspace of the Johnson scheme $J(N, F)$ (Appendix A.2). Verified: brute-force projection of the 400×400 superoperator at $N = 6$ agrees to 10^{-9} .

5.3 Exact sum rules; freeness is flat transmission [derived + verified]

$$\phi_0 = a, \quad \sum_k \phi_k \dim V_k = d^2, \quad r|_{\text{half filling, orbit-averaged}} = \frac{\phi_1}{a} \quad (13)$$

All three hold per realization to $\leq 5 \times 10^{-13}$ for $N' = 8-14$. The third uses the fact that $U = 1 - 2n_i$ has sizes 0 and 1 only, and it agrees with (9): two independent exact routes to the orbit-averaged variance.

- **Haar.** $\phi_{k \geq 1} = a(aD^2 - 1)/(D^2 - 1) = a^2 + O(D^{-2})$ is flat, and (11) gives $2a^2 - a^3$, the free value.
- **SYK.** The transmission excess $\delta\phi_k = \phi_k - a^2$ is positive at small k (Fig. 4b; $\delta\phi_1/a^2 = +50\%$ at $N' = 14$). By the second sum rule, the dimension-weighted mean of $\delta\phi$ is essentially zero.

Since $\phi_0 = w_0 = a$,

$$E \tau(P_1 P_2 P_1 P_2) = (2a^2 - a^3) + \sum_{k \geq 1} \delta\phi_k w_k. \quad (14)$$

Freeness of two BPS spaces is exactly the statement $\delta\phi \equiv 0$. SYK violates it because its size spectrum and its transmission excess sit at the same small sizes.

5.4 Two independent SYK models: prediction vs measurement [numerical]

N'	predicted (from w_k)	measured (pairs)	pred./free	meas./free	pairs
10	0.56399	0.56406 ± 0.00014	1.0056	1.0057	16
12	0.41606	0.41613 ± 0.00149 (plain)	1.0203	1.0205	4
		0.41674 ± 0.00030 (c.v.)		1.0219	
13	0.29481	—	1.0370	—	—
14	0.29661	0.29664 ± 0.00014	1.0434	1.0435	16
Haar	$2a^2 - a^3$	agrees	1.0000	1.000	1 per N'

Table 4: [numerical] Fourth moment of the principal-angle law between two independent SYK BPS spaces (ν = squared cosines of the principal angles). At $N' = 10, 14$ the measured value uses a control variate on $\nu - a$, whose expectation is exactly zero. At $N' = 12$ (only 4 pairs) both the plain mean and the control-variate mean are shown. The latter is 0.16% (2σ) above the prediction, and its regression coefficient is poorly determined with 4 pairs. Elsewhere the parameter-free prediction agrees to $\leq 0.02\%$. The prediction uses independent seeds (0–3) from the pairs.

6 The decoder's fourth moment

6.1 Exact decomposition [derived + verified]

Write $P' = a\mathbb{1} + Y$ with $Y = U(P - a)U$, the rotated nonzero-size part (the reflection fixes the identity component). With $P^2 = P$,

$$T_4 = \tau(PP'PP') = \underbrace{a^2}_{\text{no } P' \text{ nontrivial}} + \underbrace{2a(T_2 - a)}_{\text{one}} + \underbrace{Q_4}_{\text{both}}, \quad Q_4 = \frac{\text{Tr}(PYPY)}{d}. \quad (15)$$

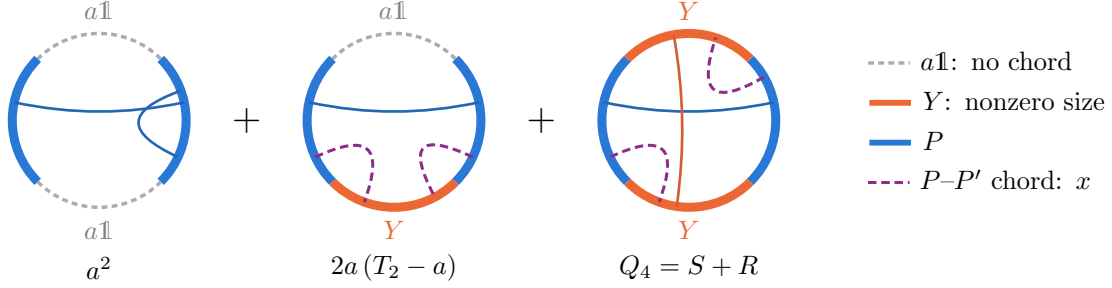


Figure 5: **[derived + verified]** + **[interpretation]** Exact decomposition of the decoder’s fourth moment $T_4 = \tau(PP'PP')$ by how many P' -arcs carry nonzero size. Write $P' = a\mathbb{1} + Y$ on each P' -arc. The identity part (dashed grey) is the zero-length wormhole: no chord attaches to it. Y (solid orange) is the nonzero-size part. The first two terms are fixed exactly by a and the variance. The third splits into S , the twirled (“independent-like”) part, exact from the size spectrum w_k , and a genuinely four-point remainder R , the correlation between Y and P beyond the twirl.

Splitting $X \mapsto PXP$ into its $U(N)$ twirl (the ϕ_k) and a remainder $\delta\mathcal{P}$,

$$Q_4 = S + R, \quad S = \sum_{k \geq 1} \phi_k w_k, \quad R = \frac{\langle Y, \delta\mathcal{P} Y \rangle}{d}. \quad (16)$$

- This is exact per realization and for any probe U .
- Free (Haar): $T_2 = a$, $S = a^2(1 - a)$, $R = 0$, so $T_4 = 2a^2 - a^3$. Note that the free T_4 already contains $a^2(1 - a)$ from nonzero sizes; this is the correction to yesterday’s conjecture (item 9 of the summary).
- $R(\mathbb{1}) = (1 - a)^2 - S$. For a Haar-random probe $g \in U(N)$, $E_g R = 0$ and $E_g T_2 = a$.

Figure 5 shows the three terms as chord pictures.

6.2 SYK numbers **[numerical]**

N'	a	T_2	T_4	T_4/T_4^{free}	$2a(T_2 - T_2^{\text{free}})$	$S - Q_4^{\text{free}}$	R	shares of excess
12	0.526	0.689	0.5994	1.470	+0.1715	+0.0083	+0.0119	89 % / 4 % / 6 %
13	0.425	0.615	0.5063	1.750	+0.1587	+0.0084	+0.0500	73 % / 4 % / 23 %
14	0.425	0.642	0.5342	1.879	+0.1845	+0.0123	+0.0531	74 % / 5 % / 21 %
Haar		$\approx a$		1.000 ± 0.001	≈ 0	0	≈ 0	–

Table 5: **[numerical]** Decoder fourth moment: mean over 4 realizations $\times$ 5–6 modes. Free baseline: Wachter(a, b) moments of μ ($T_2^{\text{free}} = a$, $Q_4^{\text{free}} = a^2(1 - a)$ at even N'). The shares add the three pieces of $T_4 - T_4^{\text{free}}$. Not shown: $N' = 8, 10$ (2 realizations), where $a > 0.618$ makes $R(\mathbb{1}) < 0$. There R is negative and the variance term alone exceeds the total excess.

6.3 What is known about R

- **[derived]** The exact limits are $R(\mathbb{1}) = (1 - a)^2 - S$ and $E_{g \in U(N)} R = 0$. R is the only piece that needs information beyond the size spectrum: as a class function of the probe, it involves four-copy moments of the ensemble.

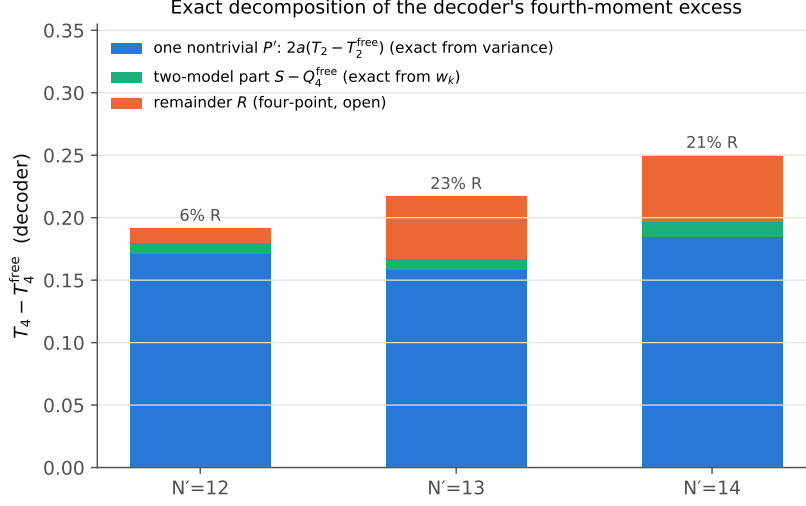


Figure 6: [numerical] The decoder’s fourth-moment excess over free, split exactly: the variance term (blue), the two-model size-spectrum term (green) and the four-point remainder R (orange). About 77–79% at $N' = 13$ –14 (94% at $N' = 12$) is fixed exactly by the size spectrum and the variance.

- [numerical] Two natural closures fail.
 - Size-diagonal characters, $R \approx \sum_{k,k'} \hat{\chi}_k \hat{\chi}_{k'} \langle P^{(k)}, \delta \mathcal{P} P^{(k')} \rangle / d$, replace the reflection’s character on each two-copy irrep in $V_k \otimes V_{k'}$ by its average. This has the right sign at $N' = 10, 12$ but is 2–5 $\times$ too small (0.0055 vs 0.0117 at $N' = 12$).
 - A “coherent mixture” $Y \approx \eta P_{\geq 1} + \sqrt{1 - \eta^2} Y_{\text{indep}}$, with $\eta = (T_2 - a)/(1 - a)$, predicts $R = \eta^2 R(1)$, i.e. $T_4 \approx T_2^2 + (1 - \eta^2)S$. It is good at $N' = 12$ (mean -0.3%) but systematically 4–5% low at $N' = 13, 14$.
- [numerical] Empirically $R/R(1) = 0.23$ –0.25 at $N' = 13$ –14. This lies between $\eta^2 \approx 0.11$ –0.14 (the mixture ansatz) and $\eta \approx 0.33$ –0.38. R is thus more strongly correlated with the variance than an independent-noise ansatz allows. This is an observation, not a law: the single-site probes cluster at few values of η .

7 Toward a gravity dual of the decoder

7.1 Kinematics versus dynamics

Every exact identity in §§4–6 follows from $U(N)$ invariance and representation theory. They hold for *any* $U(N)$ -invariant ensemble of projectors. The model-specific content, which is what a gravity dual must supply, is concentrated in a few spectra:

1. the size spectrum w_k of the BPS projector, which fixes the variance exactly, eq. (9);
2. nothing more for the two-model fourth moment, since ϕ_k is a universal function of w_k , eq. (11);
3. four-copy data for the decoder’s remainder R .

This sharpens the target. *A gravity dual of the decoder must, first of all, predict the operator-size spectrum of the BPS projector.* The super-chord theory does so approximately: $w_{2n} \approx P_n$ (Table 3). SYK serves as the exemplar against which such predictions are tested.

7.2 Dictionary so far

Table 6 collects the correspondences between exact finite- N objects and chord quantities. The status column separates exact statements from empirical matches.

exact finite- N object	chord / gravity counterpart	status
identity component $P^{(0)} = a\mathbb{1}$, $w_0 = a$	zero-length SUSY wormhole, P_0	exact both; $a_\lambda \neq a$ at $p = 3$
size spectrum w_{2n}	wormhole-length distribution P_n	empirical, 10–20 %
character $\hat{\chi}_k$ of the probe	matter propagator $x^{2n} = q^{2\Delta n}$	structural analogy
$r = a + \sum_{k \geq 1} w_k \hat{\chi}_k$	$r = \sum_n P_n x^{2n}$ (BLY 4.5–4.8)	exact / 2 %
transmission ϕ_k	zero- T two-point of matter with $\Delta_k = k/p$	$k \leq 2$ close at $N' = 12$; $k \geq 3$ deviates
P - and P' -arcs; mixed chords cost x	two supercharges, correlation x per chord	exact per chord
$\sum_k \delta\phi_k w_k > 0$ (two models)	crossings between two chord families	exact / [interpretation]
decoder R (both P' nontrivial)	four-arc diagrams with mixed chords	open
freeness	flat transmission, $x = 0$	exact characterization

Table 6: Correspondences established or suggested so far.

7.3 Next computations

1. **Derive $n \leftrightarrow k/2$.** Relate super-chord number to operator size, using the operator-size content of BLY’s n_O, n_X (their §5.2, eqs. 5.20–5.23). A derivation would promote Table 3 from a match to a dictionary entry.
2. **The four-arc chord computation.** Build the correlated two-supercharge chord Hilbert space: BLY’s §5 machinery with a $\mathbb{Z}_2$ flag per chord that records whether it straddles a U . Compute $T_4(x)$, then $R(x)$ and the two-model crossing term.
3. **Two-copy content of R .** Characters of the reflection on the irreps of $V_k \otimes V_{k'}$ carrying R (Casimir on superoperators; feasible for $N' \leq 12$).

8 Open questions and dubious points

Items that did not pass the audit as established facts, or that are known limitations:

1. [conjecture] $n \leftrightarrow k/2$. The size- $2n$ weights match P_n to 10–20 %. It is not derived why one super-chord “pair” carries size 2 rather than, say, p .
2. [conjecture] **Why the closures for R fail.** The proposed reason is that R ’s weight sits on *small* two-copy irreps, where the reflection acts almost trivially. This was inferred, not measured.
3. **Chord factorization at finite N .** The $\mathbb{Z}_2$ weight is exact per chord, but independence between chords is a double-scaled approximation. The effective per-chord weight at our sizes lies between the exact $(1 - 2p/N')$ and double-scaled $(e^{-2p/N'})$ values. Inverting $T_2(x_s)$ for P_n gives $P_1 > 1$ with the former; with the latter the intermediate- s prediction is 5–10 % high. For T_2 the exact sum rule of §4 bypasses the issue. For a chord computation of T_4 it remains.

4. **[interpretation]** “ $x = 0 = \text{independent models} = \text{Haar-random probe}.$ ” The exact content is $E_g T_4(g) = \sum_k \phi_k w_k$ and $E_g T_2 = a$ for g Haar in $U(N)$, per realization; the equality with independent models holds in expectation. The parity probe at $s = N'/2$ is close to but slightly more non-free than two independent models (T_4 : 1.056 vs 1.043 at $N' = 14$), consistent with its small nonzero characters.
5. **Even sizes at half filling:** numerically exact, but the particle–hole derivation is a sketch with signs not tracked.
6. **Odd N' .** The free baseline must use Wachter(a, b) with $b \neq \frac{1}{2}$ (done here). The mixture ansatz and the R analysis have not been generalized.
7. **Asymptotics.** Everything is at $N' \leq 14$ (≤ 16 for r). The x -independent non-freeness grows with N' ($0.6 \rightarrow 4.3\%$), and R ’s share grows ($6 \rightarrow 21\%$). Their large- N' fate, including the difference between double scaling and fixed $p = 3$, is unknown.
8. **Statistics.** $N' = 14$ uses 4 realizations for sizes and T_4 . The parity-family sum rule holds within 2.5σ , with tensions of $+2.4\sigma$ ($N' = 14, s = 7$) and -2.0σ ($N' = 13, s = 3$).
9. **The density.** Only moments through T_6 are studied. The spectral density itself has not been reconstructed or predicted.
10. **Superseded:** “the free (Wachter) law is the zero-length truncation of the chord computation.” This is true for m_2 only (§6); the working note’s “falsified in strong form” is replaced by the exact statement above.

A Derivation details

A.1 The reflection’s characters

Averaging $U_v = 1 - 2n_v$ over its $U(N)$ orbit means $n_v = \sum_{ab} v_a \bar{v}_b E_{ab}$ with v a Haar-random unit vector. Haar moments give $E_v[n_v] = \hat{N}/N = F/N$ and

$$E_v[n_v X n_v] = \frac{\hat{N} X \hat{N} + \sum_{ab} E_{ab} X E_{ba}}{N(N+1)}.$$

Write $\sum_{ab} E_{ab} X E_{ba} = c_F X - \frac{1}{2} C(X)$, with $c_F = \sum_{ab} E_{ab} E_{ba}|_{\Lambda^F} = F(N+1-F)$ (checked numerically). Then

$$E_v[U_v X U_v] = X \left[1 - \frac{4F}{N} + \frac{4(F^2 + c_F)}{N(N+1)} \right] - \frac{2}{N(N+1)} C(X) = X - \frac{2}{N(N+1)} C(X).$$

On V_k , $C = 2k(N+1-k)$, which gives $\hat{\chi}_k(U) = 1 - 4k(N+1-k)/(N(N+1))$. Checked against the Krawtchouk form to 10^{-16} for $N = 7 \dots 20$.

A.2 The transmission matrix

$\text{Tr}[\mathcal{R}(Y) \text{Ad}_g] = \langle \langle \rho(g)^\dagger | Y | \rho(g)^\dagger \rangle \rangle$. So column k of $\mathcal{A}$ is the character expansion of the class function $g \mapsto \|\rho(g)^{(k)}\|^2$. On the torus $\rho(g) = \text{diag}(z^S)$, and its size- k part is the projection onto the k -th Johnson eigenspace (S_N irrep $(N-k, k)$, the weight-zero space of V_k), with entries $e_k(r)$ depending on $r = |S \setminus T|$. Using $\sum_{|S \setminus T|=r} z^S \bar{z}^T = \binom{N-2r}{F-r} m_r$ and $\chi_{V_k} = \sum_r \left[\binom{N-2r}{k-r} - \binom{N-2r}{k-r-1} \right] m_r$ gives a triangular system for each column.

A.3 Haar transmission

For Haar P of rank d , the Weingarten formula gives $E[PPX] = \frac{d(dD-1)}{D(D^2-1)}X$ on traceless X . So $\phi_{k \geq 1} = a(aD^2 - 1)/(D^2 - 1)$, and $\sum_k \phi_k w_k = 2a^2 - a^3 + O(D^{-2})$.

B Reproducibility

All numbers are exact linear algebra; every heavy run was under the memory watchdog (peak ≤ 1.0 GB).

Item	Code / data
Parity family (Fig. 3, Tab. 1)	scripts/run_parity_family.py, scripts/analyze_parity_family.py; results/data/parity_family_2026-09-29/raw_q3_N*.jsonl, summary.csv
Size spectrum, sum rule (Tab. 2, 3)	src/size_decomposition.py, scripts/run_size_decomposition.py, scripts/analyze_size_sumrule.py; size_q3_N*.jsonl
Transmission, two models (Fig. 4b, Tab. 4)	scripts/run_two_model_overlap.py, scripts/analyze_two_copy.py; two_model_q3_N*.jsonl, two_copy_tables.md
Decoder T_4 (Tab. 5, Fig. 6)	scripts/run_decoder_T4.py, scripts/analyze_decoder_T4.py; decoderT4_q3_N*.jsonl, decoderT4_tables.md
Figures	scripts/plot_progress_2026-09-29.py $\rightarrow$ results/figures/progress_2026-09-29/
Tests	tests/test_size_decomposition.py (Casimir spectrum, characters, $w_0 = a$, Haar sum rule, brute-force transmissions, exact T_4 decomposition)
Working note and audit	research/notes/decoder_moments_2026-09-29.md, research/notes/decoder_moments_audit_2026-09-29.md

Realizations and seeds. $p = 3$, $F = \lfloor N'/2 \rfloor$ throughout.

- SYK coupling seeds 0–3 for the parity family ($N' = 8$ –14), the size spectra ($N' = 8$ –14) and the decoder T_4 at $N' = 12$ –14. The decoder T_4 at $N' = 8, 10$ uses seeds 0–1.
- Independent pairs use seeds (100, 101), $\dots$, (106, 107), plus (110, 111), $\dots$, (132, 133) at $N' = 10, 14$.
- Haar controls: seeds 0–1 (parity family; seed 0 at $N' = 13$), seed 0 (sizes, T_4), pair (200, 201).
- Single-site probes: 4–6 modes per realization for T_4 .
- Data were generated from commit 86fa66e plus the working tree (recorded in `meta_*.json`). The report is at 774934a plus the uncommitted audit, figure script and this file.

References

- [1] J. Chryssanthacopoulos and D. Vegh, *Fortuity and fragility in supersymmetric SYK*, arXiv:2608.12160 (2026).
- [2] J. Boruch, H. W. Lin and C. Yan, *Exploring supersymmetric wormholes in $\mathcal{N} = 2$ SYK with chords*, arXiv:2308.16283 (2023).
- [3] H. W. Lin, J. Maldacena, L. Rozenberg and J. Shan, *Looking at supersymmetric black holes for a very long time*, SciPost Phys. **14** (2023) 128, arXiv:2207.00408.

- [4] W. Fu, D. Gaiotto, J. Maldacena and S. Sachdev, *Supersymmetric Sachdev-Ye-Kitaev models*, Phys. Rev. D **95** (2017) 026009, arXiv:1610.08917.
- [5] A. Nica and R. Speicher, *Lectures on the Combinatorics of Free Probability*, London Math. Soc. Lecture Note Series 335, Cambridge University Press (2006).
- [6] B. Collins, S. Matsumoto and J. Novak, *The Weingarten Calculus*, arXiv:2109.14890 (2021).